7.7.1 Expansion of the Operator $\hat{\mathcal{L}}$

Consider a Langevin splitting method using the A, B and O splitting pieces, where we compose exact solves of each of the vector fields in a prescribed sequence to define our discretization scheme. Because we solve each piece exactly, we are able to write down the action of each individual solve on the initial distribution in terms of the vector field's evolution operator (using the respective generators for each piece as in [99]), where

$$\begin{aligned}
\left(e^{t\mathcal{L}_{\mathrm{A}}}\phi\right)(\boldsymbol{q},\boldsymbol{p}) &= \phi(\boldsymbol{q}+t\boldsymbol{M}^{-1}\boldsymbol{p},\boldsymbol{p}),\\
\left(e^{t\mathcal{L}_{\mathrm{B}}}\phi\right)(\boldsymbol{q},\boldsymbol{p}) &= \phi(\boldsymbol{q},\boldsymbol{p}-t\nabla U(\boldsymbol{q})),\\
\left(e^{t\mathcal{L}_{\mathrm{O}}}\phi\right)(\boldsymbol{q},\boldsymbol{p}) &= \int_{\mathcal{P}} \phi\left(\boldsymbol{q}, e^{-\gamma t}\boldsymbol{p}+\zeta\boldsymbol{M}^{1/2}\boldsymbol{x}\right)\frac{e^{-|\boldsymbol{x}|^2/2}}{(2\pi)^{\mathrm{N_c}/2}}\mathrm{d}\boldsymbol{x},
\end{aligned}$$

where $\zeta = \left[\mathrm{k_B T}\left(1-e^{-2\gamma t}\right)\right]^{1/2}$. The overall generator for an associated Langevin splitting method $\hat{\mathcal{L}}$ can be computed by composing each individual operator associated to the terms in the splitting. For example, for the method labelled by the string $[\![\mathrm{ABAO}]\!]$, we have

$$\exp(h\hat{\mathcal{L}}_{[\![\mathrm{ABAO}]\!]}) = \exp((h/2)\mathcal{L}_{\mathrm{A}})\exp(h\mathcal{L}_{\mathrm{B}})\exp((h/2)\mathcal{L}_{\mathrm{A}})\exp(h\mathcal{L}_{\mathrm{O}}),$$

where importantly note the apparent reversal of the sequence of products, known as the *vertauschungssatz* (see the discussion in [164, Section III.5.1]), which is a consequence of the relation

$$\phi(\mathscr{U}_t^{\mathrm{f}_1}(z)) = \left(e^{t\mathcal{L}_{\mathrm{f}_1}}\phi\right)(z),$$

for some vector field f_1 with associated flow map $\mathscr{U}_t^{\mathrm{f}_1}$ and generator $\mathcal{L}_{\mathrm{f}_1}$, where the choice of $\phi(z) = \mathscr{U}_t^{\mathrm{f}_2}(z) = (e^{t\mathcal{L}_{\mathrm{f}_2}}\mathrm{Id})(z)$ demonstrates this reversal.

We will find it more convenient to work with the adjoint of the perturbed generator, denoted $\hat{\mathcal{L}}^\dagger$, which we may compute as

$$\exp(h\hat{\mathcal{L}}^\dagger_{[\![\mathrm{ABAO}]\!]}) = \exp(h\mathcal{L}^\dagger_{\mathrm{O}})\exp((h/2)\mathcal{L}^\dagger_{\mathrm{A}})\exp(h\mathcal{L}^\dagger_{\mathrm{B}})\exp((h/2)\mathcal{L}^\dagger_{\mathrm{A}}).$$

Under the propagation of the numerical scheme (or the discretized dynamics), the Fokker-Planck equation gives us that the evolution of a density of points ρ is given by

$$\frac{\partial}{\partial t}\rho(z,t) = \hat{\mathcal{L}}^\dagger\rho,$$